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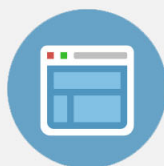
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Classical trajectory and statistical adiabatic channel study of the dynamics of capture and unimolecular bond fission. V. Valence interactions between two linear rotors

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The combination of two linear rotors forming linear or nonlinear adducts is treated using standardized valence potentials. Classical trajectory (CT) and statistical adiabatic channel (SACM) calculations are used for the calculation of thermal capture rate constants. At very low temperatures, only SACM applies. At intermediate temperatures SACM and CT approach each other; however, Landau–Zener-type multiple crossings of adiabatic channel potentials introduce local nonadiabaticity which has to be accounted for. The high-temperature transition from globally adiabatic to nonadiabatic (sudden) dynamics is studied by CT. Thermal rigidity factors, accounting for the influence of the anisotropy of the potential on the capture rate constant, are expressed in simple analytical form which facilitates practical applications. The present work complements similar studies on the addition of atoms to linear molecules in standardized valence potentials (part IV of this series). © 1998 American Institute of Physics. [S0021-9606(98)00924-6]

I. INTRODUCTION

In part IV of this series¹ we performed extensive classical trajectory (CT) and statistical adiabatic channel (SACM) calculations of thermal rate constants for the capture of atoms by linear species interacting in simple valence potentials. It was possible to represent the results in compact analytical form as a function of a few molecular parameters. In addition, the transition between classical adiabatic and nonadiabatic (sudden) dynamics was characterized. Part IV continued a series of earlier investigations for the capture of ions by linear dipoles (part I²) and by quadrupoles (part II³) as well as of linear dipoles by linear dipoles (part III⁴). Part IV replaced the electrostatic potentials of parts I–III by simple valence potentials with Morse attraction and simple anisotropy models.⁵

It was shown in parts I–IV that the conditions for “global” adiabatic dynamics were fulfilled for most conditions of practical interest. However, the dipole–dipole systems showed some differences to systems where atoms were captured: it was found^{6–8} that the combination of two di- or polyatomic species, except for the lowest adiabatic channels, shows numerous simple and complex avoided crossings between the channel potential curves of the Landau–Zener type. For these crossings, transitions between adiabatic and “locally” nonadiabatic dynamics in general cannot be neglected. As the problem becomes very complicated on the SACM level, except for very low temperatures, the CT treat-

ment may replace the SACM treatment, or, at least, nonadiabatic corrections to the SACM have to be designed. The SACM treatment, nevertheless, provides the only solution for lower temperatures where CT break down. Similar to part IV, where the ion–dipole treatment was modified to apply to a valence potential between an atom and a linear species, in the present work we modify the dipole–dipole potential in order to calculate thermal capture rate constants for a valence potential between two linear species. Again the combination of high-temperature CT and low-temperature SACM calculations provides an access to explicit analytical relations between capture rate constants and molecular parameters.

Of particular interest is a representation of the rate constants in terms of an expression from phase-space theory (PST), i.e., for an isotropic potential, and a thermal rigidity factor $f_{\text{rigid}}(T)$ accounting for the anisotropy. In part IV, f_{rigid} was related in a simple way to the ratio $C = \epsilon_e^2/2BD$ with the rotational constant B of the linear species, the Morse dissociation energy D of the adduct, and its bending frequency ϵ_e in the equilibrium position, as well as to the ratio α/β between the “looseness parameter” α and the Morse parameter β of the potential.⁵ It arises the question whether, for the linear rotor+linear rotor system of the present work, analogous relations can be found such as suggested by the much cruder treatment in Ref. 9. The obtained results will serve as a reference for comparison with calculations using more complicated individual valence potentials (see, e.g., the potential for the reaction $\text{OH} + \text{OH} \rightleftharpoons \text{H}_2\text{O}_2$ in Ref. 10). Apart from the general insight provided by simple expressions for $f_{\text{rigid}}(T)$, their practical usefulness is quite evident.

In the present article we do not give an overview of the literature on PST, CT, or SACM calculations. Some of this has been included in parts I–IV.^{1–4} Instead, like in part IV, we directly proceed to a systematic study of the influence of molecular parameters on thermal capture rate constants.

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II. MODEL POTENTIALS, REDUCED HAMILTONIANS, AND MASSEY PARAMETERS

The present SACM and CT calculations are based on two sets of standardized valence potentials. At first, we consider the formation of linear adducts from two linear species interacting on potentials of the form

$$V(r, \text{angles}) = -2D \exp[-\beta(r-r_e)] + D \times \exp[-2\beta(r-r_e)] + A_1 \exp[-2\alpha(r-r_e)] \times \{2 + [\mathbf{d}_1 \mathbf{d}_2 - 3(\mathbf{d}_1 \mathbf{n})(\mathbf{d}_2 \mathbf{n})]\}/4 \quad (2.1)$$

with the parameters D , β , and r_e of the Morse potential, and the “looseness parameter” α of the anisotropy of the potential (see the original version of the SACM⁵). r is the distance between the centers of mass of the two combining species. The last term in Eq. (2.1) characterizes the anisotropy; it is chosen to be of the form of the dipole–dipole interaction, however, with a different “anisotropy amplitude.” $\mathbf{n}$ denotes the unit vector along the line connecting the two centers of mass, $\mathbf{d}_1$ and $\mathbf{d}_2$ are the unit vectors along the axes of the two linear rotors. The “anisotropy amplitude” of the last term in Eq. (2.1) and its parameter A_1 in a simple way are related to the quanta of the deformation vibrations of the four “transitional modes” of the linear adduct, see below. Using the potential of Eq. (2.1), the Hamiltonian of the system is written as

$$H = \frac{J_1^2}{2I_1} + \frac{J_2^2}{2I_2} + \frac{p^2}{2\mu} + V(r, \text{angles}), \quad (2.2)$$

where J_1 and J_2 are the angular momenta of the two combining linear rotors, I_1 and I_2 are their moments of inertia, p is the linear momentum of the relative motion, and μ is the reduced mass of the two rotors. Our calculations were done for identical rotors; for different rotors $B = \hbar^2/2I = (B_1 + B_2)/2$ defines the effective rotational constant of the system with $I = 2I_1 I_2 / (I_1 + I_2)$. The Hamiltonian of Eqs. (2.1) and (2.2), at not too large values of r , correspond to two doubly degenerate deformation vibrations⁶ with the quanta

$$\epsilon_{AS}(r) = \epsilon_{AS}(r_e) \exp[-\alpha(r-r_e)], \quad (2.3)$$

$$\epsilon_S(r) = \epsilon_S(r_e) \exp[-\alpha(r-r_e)],$$

which are related to A_1 from Eq. (2.1) by

$$\epsilon_{AS}(r_e) = \hbar \sqrt{A_1/4I}, \quad (2.4)$$

$$\epsilon_S(r_e) = \hbar \sqrt{3A_1/4I}$$

[see Eq. (15) of Ref. 6]. The relation

$$A_1 \approx 2[\epsilon(r_e)]^2/B\sqrt{3} \quad (2.5)$$

with

$$\epsilon(R_e) = \sqrt{\epsilon_{AS}(r_e) \epsilon_S(r_e)} \quad (2.6)$$

thus provides the link between the “anisotropy amplitude” in Eq. (2.1) and the vibrational quanta of the two doubly degenerate bending vibrations of the linear adduct in the equilibrium configuration at $r=r_e$. [It should be noted that $\mu r_e^2 \gg I$ is assumed in Eq. (2.4) by analogy to Eq. (2.4) from part IV,¹ see also below.]

Equation (2.1) characterizes the formation of a linear adduct with an anisotropy of the potential of the dipole–dipole type; however, the attraction potential and the anisotropy amplitude are chosen to be of standardized valence character. In addition to this potential, we have also studied a model potential which leads to nonlinear adducts. We considered a potential of the type

$$V(r, \theta_1, \varphi_1, \theta_2, \varphi_2) = -2D \exp[-\beta(r-r_e)] + D \times \exp[-2\beta(r-r_e)] + A_2 \times \exp[-2\alpha(r-r_e)] U(\theta_1, \varphi_1, \theta_2, \varphi_2), \quad (2.7)$$

where φ_1 , θ_1 , φ_2 , and θ_2 are the spherical coordinates of the two rotors, the line connecting the centers of mass being the polar axis. The function $U(\theta_1, \varphi_1, \theta_2, \varphi_2)$ was constructed to be the difference between some function U_a and its angular minimum value $U_{a,\min}$, i.e.,

$$U = U_a - U_{a,\min}. \quad (2.8)$$

U_a was expressed by

$$U_a = -\cos \varphi \sin \theta_1 \sin \theta_2 - \cos \theta_1 \cos \theta_2 + \alpha^* \cos \theta_1 \cos \theta_2 + \beta^* (\cos \theta_1 - \cos \theta_2) + \gamma^* (\cos^2 \theta_1 + \cos^2 \theta_2)/2, \quad (2.9)$$

where $\varphi = \varphi_1 - \varphi_2$. U_a can likewise be written as

$$U_a = -\mathbf{d}_1 \mathbf{d}_2 + \alpha^* (\mathbf{d}_1 \mathbf{n})(\mathbf{d}_2 \mathbf{n}) + \beta^* \{(\mathbf{d}_1 \mathbf{n}) - (\mathbf{d}_2 \mathbf{n})\} + \gamma^* \{(\mathbf{d}_1 \mathbf{n})^2 + (\mathbf{d}_2 \mathbf{n})^2\}/2, \quad (2.10)$$

where $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{n}$ have the same meaning as in Eq. (2.1). As long as $2 + \gamma^* - \alpha^* > 0$, $U_{a,\min}$ is obtained for the symmetric configuration $\varphi = 0$, $\theta_1 = \pi - \theta_2 = \theta$ with $\cos \theta = -\beta^*/(2 + \gamma^* - \alpha^*)$. θ thus is the angle between the axis of the linear rotors and the line connecting their centers of mass. The parameters α^* , β^* , γ^* , and $U_{a,\min}$ can be related to the angle θ and the frequency-related quantities Ω_T , Ω_S , and Ω_{AS} for torsion, symmetric, and asymmetric deformation vibrations. With

$$\chi_{AS}^2 = (\Omega_{AS}/\Omega_T)^2 \quad (2.11)$$

and

$$\chi_S^2 = (\Omega_S/\Omega_T)^2 \quad (2.12)$$

one has

$$\alpha^* = 2 + (\chi_{AS}^2 - \chi_S^2 - 1)/\sin^2 \theta, \quad (2.13)$$

$$\beta^* = -2\chi_S^2 \cos \theta / \sin^2 \theta, \quad (2.14)$$

$$\gamma^* = (\chi_{AS}^2 + \chi_S^2 - 1)/\sin^2 \theta. \quad (2.15)$$

From the frequencies $\omega_T(r=r_e)$, $\omega_S(r=r_e)$, and $\omega_{AS}(r=r_e)$ of the torsion, symmetric, and asymmetric deformation vibrations and the angle θ , the quantities A_2 , α^* , β^* , and γ^* are constructed. One has

$$\Omega_T = \sqrt{2A_2/I} = \omega_T(r=r_e)/\sqrt{1 + 2I \cos^2 \theta / \mu r_e^2}, \quad (2.16)$$

$$\Omega_S = \omega_S(r=r_e), \quad (2.17)$$

$$\Omega_{AS} = \omega_{AS}(r=r_e)/\sqrt{1+2I/\mu r_e^2}. \quad (2.18)$$

In general,¹ $2I/\mu r_e^2 \ll 1$ can be safely assumed such that $\Omega_i \approx \omega_i(r=r_e)$. It should be noted that, for an adduct with parallel rotors, one has $\cos \theta = 0$ and $\beta^* = 0$. In this case, the interchange of χ_S^2 and χ_{AS}^2 does not change the capture rate constant. Later on, potentials of the form of Eqs. (2.7)–(2.10) are characterized by the parameters α/β , βr_e , A_2/D , and θ .

Our CT calculations as before were done for reduced quantities with a time unit $t_u = \sqrt{2I/kT} = \hbar/\sqrt{BkT}$, a length unit $l_u = 1/\beta$, an energy unit $E_u = kT$, and a mass unit $m_u = 2I/l_u^2 = 2I\beta^2$. With these units, the Hamiltonian [Eq. (2.2)] of the described potentials [Eqs. (2.1) and (2.7)] is brought into reduced form. For example, for the potential of Eq. (2.1), one has

$$h = \alpha_1^* j_1^2/2 + \alpha_2^* j_2^2/2 + p^2/2M - 2d \exp[-(\rho - \rho_e)] \\ + d \exp[-2(\rho - \rho_e)] + a_1 \exp[(-2\alpha/\beta)(\rho - \rho_e)] \\ \times \{2 + [\mathbf{d}_1 \mathbf{d}_2 - 3(\mathbf{d}_1 \mathbf{n})(\mathbf{d}_2 \mathbf{n})]\}/4, \quad (2.19)$$

where α_1^* and α_2^* are given by $\alpha_1^* = 2B_1/B$ and $\alpha_2^* = 2B_2/B$ (Ref. 6) for different rotors, being $\alpha_1^* = \alpha_2^* = 2$ for identical rotors. Analogous to part IV, j_1 and j_2 are reduced angular momenta of the linear rotors, $\rho = r/l_u = \beta r$, $\rho_e = \beta r_e$, $d = D/kT$, and $a_1 = A_1/kT$.

$$M = \mu/m_u = \mu B/\hbar^2 \beta^2 \quad (2.20)$$

defines the effective mass M of the system which is related to the relevant Massey parameter ξ of the system and which we define by⁷

$$\xi = \sqrt{2M}. \quad (2.21)$$

(This relation between ξ and M appears more appropriate than that chosen in parts I–III.) The Massey parameter, given by the product of the collision time and the circular rotational frequency, is a measure of adiabaticity, $\xi = 1$ defining the limit between globally adiabatic ($\xi > 1$) and nonadiabatic ($\xi < 1$) dynamics. The dependence of thermal capture rate constants on M will be investigated below.

Thermally averaged capture rate constants k_{cap} like in part IV are expressed by the average capture cross section $\langle \sigma \rangle$ (in reduced units) through

$$k_{\text{cap}} = \sqrt{\frac{8\pi kT}{\mu}} \frac{f}{\beta^2} \frac{\langle \sigma \rangle}{\pi}, \quad (2.22)$$

where a stoichiometric coefficient of $f = 1/2$ is used for identical rotors, or $f = 1$ otherwise. It is convenient later on to represent $\langle \sigma \rangle$, for the relevant molecular parameters, as a function of kT/D or of $kT/[D \exp(\beta r_e)]$. The same capture criteria were applied in this work as in part IV.¹

III. HARMONIC OSCILLATOR MODELS

Provided that the temperature is not too low, such that the true adiabatic channel potentials have to be considered, one may investigate a classical harmonic oscillator approximation of SACM. This model was found^{1,4} to work very well for the addition of atoms to linear rotors such that time-

consuming CT calculations could be avoided. However, for dipole–dipole capture, errors in the capture rate constants from the oscillator model of about 20% were observed in the high-temperature range which were attributed to avoided crossings of channel potentials and local nonadiabaticities. In order to correct for these effects, in the present work we elaborate an empirical correction to be implemented into oscillator versions of SACM. Before doing this, it appears useful to further inspect oscillator models with the goal to derive general relationships between capture cross sections and the characteristic parameters B , D , and $\epsilon(r_e)$.

One can employ a very similar form of the oscillator model as in part IV. In reduced form, adiabatic channel potential curves $u(v) = V_i/kT$, for the potential of Eq. (2.1) with linear adduct formation, are expressed as

$$u(v) = \lambda^2/4d - \lambda + v\sqrt{2\lambda\delta}, \quad (3.1)$$

where $\delta = a_1/16d$, $\lambda = 2d \exp[(-2\alpha/\beta)(\rho - \rho_e)]$, and $v = v_{AS}\sqrt{2} + v_S\sqrt{6}$ with the vibrational quantum numbers v_{AS} and v_S of the two degenerate oscillators. The reduced centrifugal energy, to be added to Eq. (3.1) in order to obtain the effective potential, is given by $l^2/2M\rho^2 = eb^2/\rho^2$ with the reduced angular momentum l , the reduced collision energy e , and the reduced impact parameter b . The maximum impact parameter $b(e, f)$ for capture has to be calculated numerically as a function of e and $f = v\sqrt{\delta}$ such that $b(e, f)$ can be used in the expression for the capture cross section.

In calculating the reduced capture cross section $\langle \sigma \rangle$, one has to account for the degeneracies v_{AS} and v_S of states of the two harmonic oscillators; an additional degeneracy factor of 2 accounts for the $\mathbf{d}_1 \rightarrow -\mathbf{d}_1$ and $\mathbf{d}_2 \rightarrow -\mathbf{d}_2$ symmetry.⁶ Neglecting correction factors accounting for the effects of multiple avoided crossings of the higher adiabatic channel potential curves, the integrations over v_S and v_{AS} :

$$\langle \sigma \rangle / \pi \approx 2 \int v_S dv_S \int v_{AS} dv_{AS} K(v) \quad (3.2)$$

are done with some function $K(v)$ where again $v = v_{AS}\sqrt{2} + v_S\sqrt{6}$. The calculations finally lead to

$$\langle \sigma \rangle / \pi \approx (1/36) \int_0^{v_{\text{max}}} v^3 dv \int_0^\infty eb^2(e, f) \exp(-e - e_f) de, \quad (3.3)$$

where e_f denotes entrance channel energies. Channel potential curves become repulsive for $v > v_{\text{max}}$ where

$$v_{\text{max}} = (4/3) \sqrt{d/3\delta}. \quad (3.4)$$

Introducing $f = v\sqrt{\delta}$ into Eq. (3.3) gives

$$\langle \sigma \rangle / \pi \approx (1/36\delta^2) S_0 \quad (3.5)$$

with

$$S_0 = \int_0^{f_{\text{max}}} f^3 df \int_0^\infty eb^2(e, f) \exp(-e - e_f) de. \quad (3.6)$$

Equation (3.5) implies that $\langle \sigma \rangle$ becomes proportional to $\{BD/[\epsilon(r_e)]^2\}^2$ which was anticipated in the cruder SACM version of Ref. 9. This important result from the oscillator model will be confirmed below by CT calculations. It should,

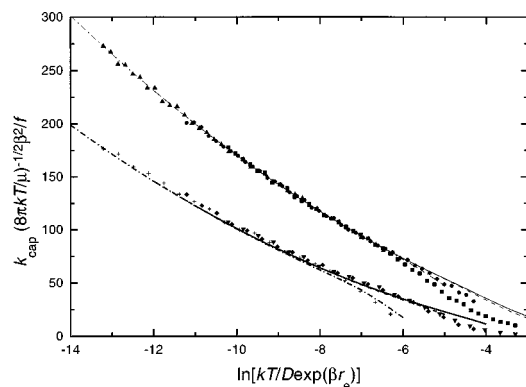


FIG. 1. Thermal capture rate constants k_{cap} for the potential of Eq. (2.1): dependence on βr_e for $A_1/D=0$ (phase-space theory, upper group of points) and for $A_1/D=4$ (lower group of points). ($\alpha/\beta=0.5$; CT results: $\blacksquare$ and $\blacktriangledown$: $\beta r_e=1$, $\bullet$ and $\blacklozenge$: $\beta r_e=2$, $\blacktriangle$ and $+$: $\beta r_e=4$; PST and SACM harmonic oscillator models: lines, from top to bottom: $\beta r_e=1,2,4$ (PST), $\beta r_e=1,2,4$ ($A_1/D=4$); with nonadiabaticity correction factor 9/8 for SACM harmonic oscillator model, see the text).

however, be emphasized that the oscillator model only applies for sufficiently strong anisotropies, see below.

IV. CAPTURE RATE CONSTANTS FROM CLASSICAL TRAJECTORY CALCULATIONS: LINEAR ADDUCTS

Reduced CT calculations were performed as in parts I–IV, studying about 10^4 trajectories for each parameter set. In the following we illustrate our results in a series of figures and by analytical fits with tabulated empirical coefficients. As in parts I–IV, we show effective capture cross-sections $\langle\sigma\rangle/\pi$, related to capture rate constants k_{cap} by Eq. (2.22), and thermal rigidity factors f_{rigid} , given by the ratio $k_{\text{cap}}/k_{\text{cap}}^{\text{PST}}$ where $k_{\text{cap}}^{\text{PST}}$ is the capture rate constant from phase-space theory, i.e., for an isotropic minimum energy path potential. At first we consider linear adducts characterized by the potential of Eq. (2.1) with $\alpha/\beta=0.5$.

Figure 1 compares k_{cap} from PST with k_{cap} for a potential with a given anisotropy. In the chosen representation, the results for different βr_e fall on one curve, except for the highest temperatures. The latter deviations are due to the effects of the short-range repulsion of the potential, see part IV.¹ The transition from $A_1/D=0$ to $A_1/D=4$ results in a decrease of the capture rate constant, i.e., it introduces “rigidity” into the system. The corresponding rigidity factors will be analyzed later. Figure 1 besides CT also includes results of the harmonic oscillator version of SACM from Sec. III corrected by an empirical curve-crossing and nonadiabaticity factor of 9/8, see below. Including this correction factor, the general agreement between CT and SACM results is very good; only at the highest temperatures some discrepancies arise which are due to slightly different capture criteria in the range of repulsive contributions of the potential.¹

Figure 2, for $\alpha/\beta=0.5$, combines results with $\beta r_e=1, 2$, and 4 and shows the dependence of k_{cap} on the strength of the anisotropy such as expressed by the ratio A_1/D . Over the range $A_1/D=0$ –8 a marked decrease of the capture rate

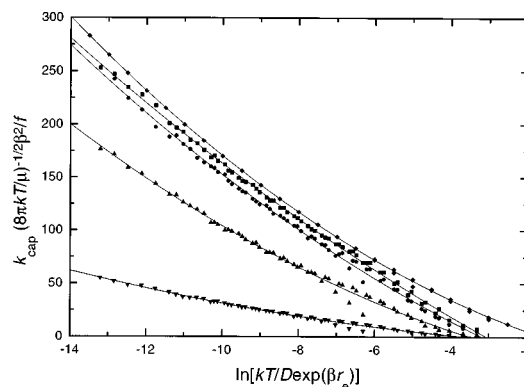


FIG. 2. Thermal capture rate constants k_{cap} for the potential of Eq. (2.1): CT results for $M=100$, $\beta r_e=1, 2$, and 4 as a function of $A_1/D=0$ ($\blacklozenge$), 1 ($\blacksquare$), 2 ($\bullet$), 4 ($\blacktriangle$), and 8 ($\blacktriangledown$) ($\alpha/\beta=0.5$; lines=fit by Eq. (4.3) with fit parameters from Table I).

constants is observed with increasing A_1/D . As the ratio A_1/D is related to characteristic molecular parameters through

$$A_1/D=4C/\sqrt{3}, \quad (4.1)$$

see Eq. (2.5), with

$$C=[\epsilon(r_e)]^2/2BD, \quad (4.2)$$

the anisotropy effects can directly be related to easily accessible molecular quantities. It should be noted that C from Eq. (4.1) has been chosen analogous to the quantity used in the addition of atoms to linear rotors, see part IV.¹ Figures 1 and 2 include fitting curves in which $Y=k_{\text{cap}}(8\pi kT/\mu)^{-1/2}\beta^2/f$ is represented as a function of $X=\ln[kT/D\exp(\beta r_e)]=\ln(kT/D)-\beta r_e$ in the form

$$Y=\alpha_1+\alpha_2X+\alpha_3X^2. \quad (4.3)$$

Except for the highest temperatures, where repulsion effects and deviations arise for different βr_e , the fits are excellent. Table I includes the fitting coefficients α_i as a function of the ratio A_1/D .

Figures 1 and 2 correspond to values of the effective mass $M=100$, i.e., to a Massey parameter ξ from Eq. (2.21) which is markedly above unity. In this case the dynamics is globally adiabatic. In order to investigate the transition from globally adiabatic to globally nonadiabatic behavior, we have reduced M down to very small values. There is practically no dependence of k_{cap} on M over the range $1\leq M\leq 100$, i.e., for $1.4\leq\xi\leq 14$. Only below $M=0.1$ ($\xi=0.45$), k_{cap} starts to decrease. Figure 3 demonstrates the transition from globally adiabatic to globally nonadiabatic (sudden) dy-

TABLE I. Fit parameters α_i in Eq. (4.3) for thermal capture rate constants: linear adduct formation with potential of Eq. (2.1) ($\alpha/\beta=0.5$).

A_1/D	α_1	α_2	α_3
0	-15.7706	-8.636	0.9975
1	-55.0049	-16.335	0.5474
2	-47.1588	-12.4674	0.7496
4	-28.7975	-5.5947	0.7684
8	-7.8000	-1.2195	0.2689

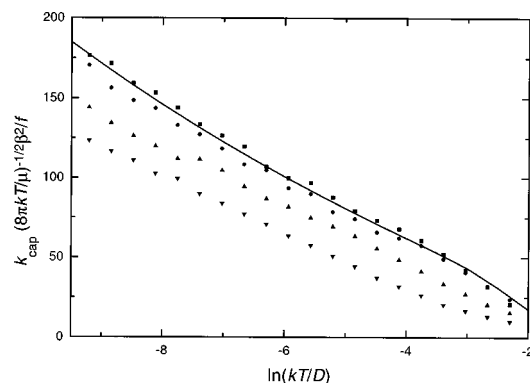


FIG. 3. Thermal capture rate constants for the potential of Eq. (2.1); transition from adiabatic to nonadiabatic (sudden) dynamics, see the text [$\beta r_e = 4$, $A_1/D = 4$, $M = 100$ (■), 0.1 (●), 0.01 (▲), 10^{-10} (▼); full line: harmonic oscillator SACM].

namics. Because global nonadiabaticity is probably of no practical relevance in thermal capture experiments, we do not further elaborate the M -dependence of k_{cap} .

Thermal rigidity factors f_{rigid} , defined by

$$k_{\text{cap}} = f_{\text{rigid}} k_{\text{cap}}^{\text{PST}}, \quad (4.4)$$

for $\alpha/\beta = 0.5$, as in part IV are temperature independent over wide ranges; only at high temperatures they decrease with increasing temperature. Figure 4 shows f_{rigid} for $\beta r_e = 1, 2$, and 4 as a function of the ratio kT/D and the anisotropy strength. The results, by analogy to part IV, can well be fitted by expressions of the form

$$f_{\text{rigid}} = \delta_1 + \delta_2 \exp(X^*/\delta_3), \quad (4.5)$$

where X^* is defined as $X^* = \ln(kT/D)$. Table II includes the fitting parameters δ_i for a series of anisotropies. The high-temperature decays of f_{rigid} are difficult to express in general form. However, the onset of the decay, for values of $X = \ln[kT/D \exp(\beta r_e)]$ below about -5 , by analogy to the results of part IV can be reasonably well reproduced by

$$f_{\text{rigid}}/f_{\text{rigid}}(T \rightarrow 0) = 1 - (A_1/D) \sqrt{\beta r_e} \exp(X/2.044). \quad (4.6)$$

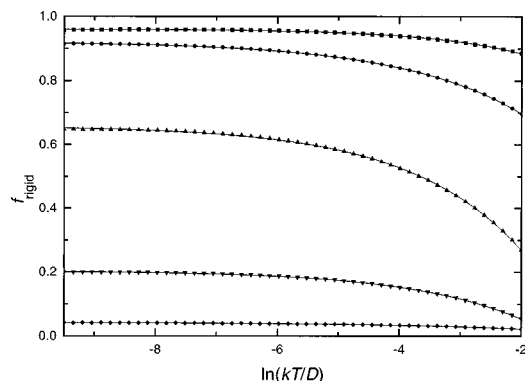


FIG. 4. Thermal rigidity factors for linear adduct formation [potential of Eq. (2.1); $\alpha/\beta = 0.5$]: temperature dependence for $\beta r_e = 4$ and $A_1/D = 1$ (■), 2 (●), 1 (▲), 0.5 (▼) [full lines: fit expressions from Eq. (4.5) and Table II].

TABLE II. Fit parameters δ_i in Eq. (4.5) for thermal rigidity factors: linear adduct formation with potential of Eq. (2.1).

βr_e	A_1/D	δ_1	δ_2	δ_3
1	1	0.9601	-4.4806	1.3117
	2	0.8984	-3.7163	1.4846
	4	0.6272	-1.8230	2.0155
	8	0.1913	-0.5208	2.1919
2	1	0.9553	-1.8595	1.2063
	2	0.9017	-1.9029	1.4726
	4	0.6365	-1.5317	1.8598
	8	0.1912	-0.5418	1.8120
4	1	0.9606	-0.2748	1.5544
	2	0.9217	-0.6206	1.9687
	4	0.6590	-1.1601	1.8375
	8	0.2033	-0.4427	1.8494
	16	0.0424	-0.0449	2.4871

[It should be noted that $f_{\text{rigid}}(T \rightarrow 0)$ does not include the quantum effects treated by SACM in Sec. VII.]

The limiting low-temperature rigidity factors depend on the parameter C , defined by Eq. (4.2), or on the ratio A_1/D in a simple way. Figure 5 shows the results. At $f_{\text{rigid}} < 0.7$, $f_{\text{rigid}}(T \rightarrow 0)$ is close to

$$f_{\text{rigid}}(T \rightarrow 0) \approx 12.5/(A_1/D)^2. \quad (4.7)$$

Over the whole range, f_{rigid} is well fitted by

$$f_{\text{rigid}}(T \rightarrow 0) = (1 + 1.5 Z_1 + Z_1^4)^{-1/4} \quad (4.8)$$

with $Z_1 = (A_1/D)^2/12.5$, such as illustrated in Fig. 5. It appears worth noticing that f_{rigid} from Eq. (4.7) (as a function of C) is nearly the square of the corresponding quantity for the addition of an atom to a linear rotor forming a linear adduct,¹ see below. We next consider cases where attraction and anisotropy of the potential do not “decrease synchronously” as in the case $\alpha/\beta = 0.5$, i.e., we study the effects of varying α/β . For $\alpha/\beta > 0.5$, the dynamics at higher temperatures samples comparably stronger anisotropies of the potential such that the rigidity factors decrease with increasing temperature (in addition to the decrease shown in Fig. 4). For $\alpha/\beta < 0.5$, the opposite is true. The situation is completely

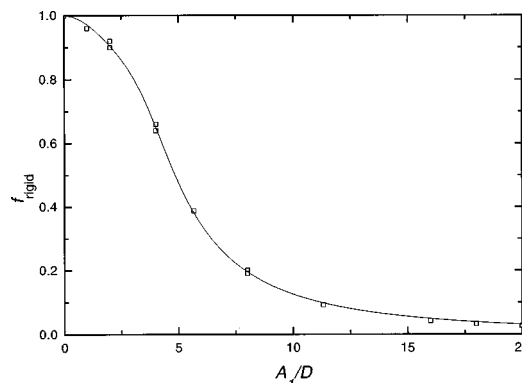


FIG. 5. Low-temperature thermal rigidity factors $f_{\text{rigid}}(T \rightarrow 0)$ for linear adduct formation [potential of Eq. (2.1); $\alpha/\beta = 0.5$]: CT results ($A_1/D \leq 8$) and harmonic oscillator SACM ($A_1/D \geq 5$; including nonadiabaticity correction factor 9/8, see the text); full line: Eq. (4.8).

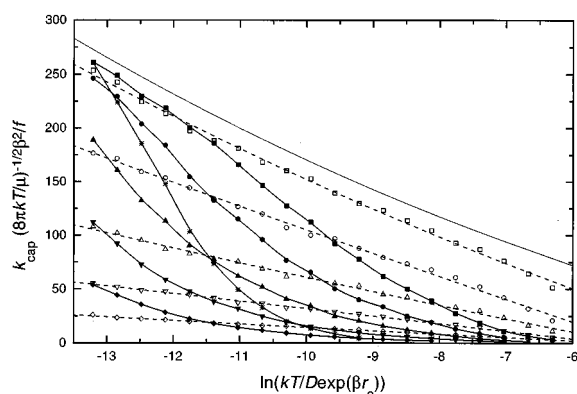


FIG. 6. Thermal capture rate constants k_{cap} for the potential of Eq. (2.1): influence of α/β . Full line on top: PST; CT for $\alpha/\beta=0.5$ with $A_1/D=2$, $\square$, 4 ($\circ$), 5.66 ($\triangle$), 8 (∇), and 11.31 ($\diamond$) like Fig. 2; CT for $\alpha/\beta=0.6$ with $A_1/D=11.31$ ($\blacksquare$), 16 ($\bullet$), 22.63 ($\blacktriangle$), 32 ($\blacktriangledown$), and 45.25 ($\blacklozenge$); CT for $\alpha/\beta=0.75$ with $A_1/D=160$ (*).

analogous to part IV. We demonstrate this behavior in Figs. 6 and 7 where k_{cap} for $\alpha/\beta > 0.5$ and for $\alpha/\beta < 0.5$, respectively, are compared with the results for $\alpha/\beta = 0.5$ from Fig. 2. As in part IV, the influence of α/β on f_{rigid} can be well accounted for by replacing C (or A_1/D) in Eqs. (4.7) and (4.8) by an effective value C_{eff} [or $(A_1/D)_{\text{eff}}$] of the form

$$C_{\text{eff}} = CF(kT/D)^{2\alpha/\beta-1} \quad (4.9)$$

with

$$F = 1 + 0.4(2\alpha/\beta - 1) + (2\alpha/\beta - 1)^2 \quad (4.10)$$

or $(A_1/D)_{\text{eff}}$ and A_1/D in Eq. (4.9).

Figure 8 demonstrates the good performance of Eqs. (4.7)–(4.10) by comparing the approximate rigidity factors with the true CT results. Only at the highest temperatures, because of the neglect of the decrease described by Eq. (4.5), are some deviations observed. These could be accounted for if necessary by modifying Eq. (4.6).

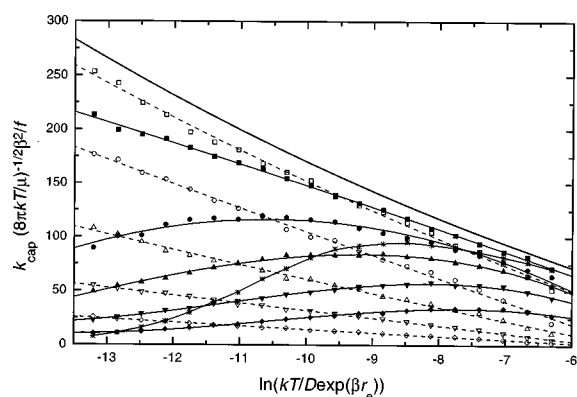


FIG. 7. Same as Fig. 6; CT for $\alpha/\beta=0.4$ with $A_1/D=0.5$ ($\blacksquare$), 1 ($\bullet$), 1.41 ($\blacktriangle$), 2 ($\blacktriangledown$), 2.83 ($\blacklozenge$); CT for $\alpha/\beta=0.25$ with $A_1/D=0.2$ (*).

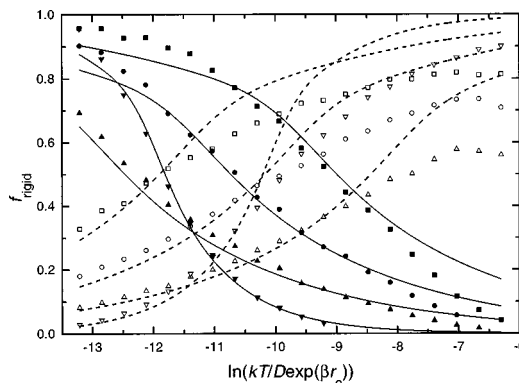


FIG. 8. Low-temperature thermal rigidity factors $f_{\text{rigid}}(T \rightarrow 0)$ for linear adduct formation: influence of α/β . CT results from $\alpha/\beta=0.25$ with $A_1/D=0.2$ (∇); $\alpha/\beta=0.4$ with $A_1/D=1$ ($\square$), 1.41 ($\circ$), 2 ($\triangle$); $\alpha/\beta=0.6$ with $A_1/D=11.31$ ($\blacksquare$), 16 ($\bullet$), 22.63 ($\blacktriangle$); $\alpha/\beta=0.75$ with $A_1/D=160$. Full and dashed lines: approximation with Eqs. (4.8)–(4.10), from top to bottom on the right-hand side: $\alpha/\beta=0.25$ with $A_1/D=0.2$, $\alpha/\beta=0.4$ with $A_1/D=1$, 1.41, 2, $\alpha/\beta=0.6$ with $A_1/D=11.31$, 16, 22.63, and $\alpha/\beta=0.75$ with $A_1/D=160$.

V. CAPTURE RATE CONSTANTS FROM CLASSICAL TRAJECTORY CALCULATIONS: NONLINEAR ADDUCTS

Besides the potential of Eq. (2.1) with its anisotropy of the dipole–dipole type leading to linear adducts, a series of calculations were done with potentials of the form of Eqs. (2.7)–(2.10) which lead to nonlinear adducts. At first, the nonlinearity results in a reduction of the number of transitional modes. In addition, in potentials of the type of Eqs. (2.7)–(2.10) increasing contributions from anharmonicity and considerable vibration–rotation couplings are encountered. This phenomenon was also observed in part IV when the anisotropy of the charge–dipole type with linear adduct formation was replaced by model potentials leading to nonlinear adducts.

The temperature dependences of the thermal capture rate constants for nonlinear adducts look quite similar to Figs. 1 and 2 and, therefore, are not reproduced here. Instead, the fit parameters of a representation of $Y = k_{\text{cap}}(8\pi kT/\mu)^{-1/2}\beta^2/f$ as a function of X^{**} [with $X^{**} = \ln(kT/D) + 4 - \beta r_e$] of the form

$$Y = \alpha_1 + \alpha_2 X^{**} + \alpha_3 X^{**2} \quad (5.1)$$

are tabulated in Table III for adducts with parallel ($\theta = 90^\circ$) and nonparallel rotors (with $\theta = 60^\circ$, 45° , and 30°).

Thermal rigidity factors f_{rigid} look quite similar to Fig. 4, i.e., with temperature-independent low-temperature values and a decrease with temperature at the highest temperatures. At first, we consider the low-temperature rigidity factors $f_{\text{rigid}}(T \rightarrow 0)$ (without accounting for quantum effects treated by SACM in Sec. VII). By analogy to Eq. (4.8) and Fig. 5, $f_{\text{rigid}}(T \rightarrow 0)$ can be represented as a function of the ratio A_2/D from Eq. (2.7). However, in addition to the ratio A_2/D , the angular properties of the potential matter are characterized by the angle θ and the frequency ratios χ_S^2 and χ_{AS}^2 from Eqs. (2.8)–(2.18). Analogous to Eq. (4.8), the best fit now is given by

TABLE III. Fit parameters α_i in Eq. (5.1) for thermal capture rate constants: nonlinear adduct formation with potential of Eqs. (2.7)–(2.10).

$\theta(^{\circ})$	χ_S^2, χ_{AS}^2	A_2/D	α_1	α_2	α_3
90	1,1	-	31.153	-18.158	0.8685
		1.0	-11.819	-24.442	0.2322
		2.0	-7.628	-10.322	0.3326
	1,4	1.0	-13.415	-17.128	0.3425
		2.0	-4.919	-5.874	0.0954
		4.0	-1.377	-1.802	0.0507
	4,4	0.5	-15.973	-22.524	0.1852
		1.1	-6.465	-7.645	0.1297
		2.4	-1.768	-2.193	0.0264
	1,1	1.0	-6.669	-18.203	0.3639
		2.2	-6.130	-8.116	0.0780
		5.0	-1.810	-2.298	0.0452
60	1,4	0.7	-2.315	-14.182	0.4852
		1.6	-4.140	-6.349	0.1480
		3.5	-1.911	-2.219	0.0325
	4,1	0.6	-3.243	-15.530	0.2642
		1.3	-5.441	-8.272	0.1703
		3.0	-2.228	-2.689	0.0344
	4,4	0.3	+3.450	-15.705	0.6352
		0.7	-6.426	-10.938	0.1105
		1.5	-3.377	-4.091	0.1015
	1,1	0.7	0.875	-13.179	0.6052
		1.5	-2.463	-7.033	0.2060
		3.0	-2.486	-3.748	0.0302
45	1,4	0.5	0.088	-11.845	0.4234
		1.0	-2.564	-6.806	0.1506
		2.0	-1.596	-3.189	0.0575
	4,1	0.2	7.602	-14.462	0.6543
		0.5	0.308	-9.398	0.1848
		1.2	-2.058	-5.022	0.0932
	4,4	0.12	2.985	-19.950	0.3805
		0.27	0.578	-11.430	0.3046
		0.9	-1.616	-4.684	0.0742
	1,1	0.3	1.621	-14.177	0.5490
		0.6	-0.132	-7.989	0.2457
		1.3	-0.881	-3.815	0.1082
30	1,4	0.15	-1.364	-19.054	0.3407
		0.3	-0.766	-9.874	0.2654
		0.9	-0.278	-3.163	0.1084
	4,1	0.09	4.221	-13.441	0.5536
		0.2	1.288	-7.725	0.2089
		0.5	1.058	-3.250	0.1425
	4,4	0.09	3.335	-10.932	0.5454
		0.2	0.213	-6.474	0.1760
		0.5	0.392	-2.672	0.1066

$f_{\text{rigid}}(T \rightarrow 0) = (1 + 0.75 Z_2 + Z_2^4)^{-1/4}$ (5.2)

with

$Z_2 = (A_2/D)^n / \gamma_1$, (5.3)

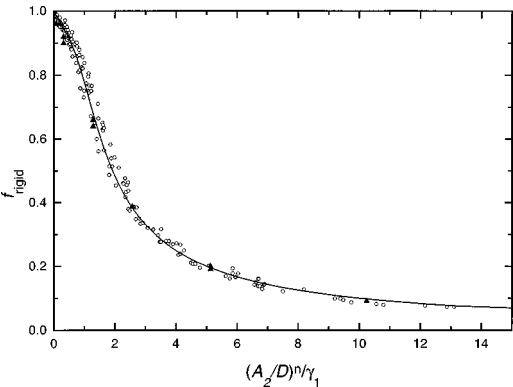


FIG. 9. Low-temperature thermal rigidity factors $f_{\text{rigid}}(T \rightarrow 0)$ for nonlinear adduct formation. $\circ$: results for examples of Table III; full line: fit with Eqs. (5.2)–(5.6), $\blacktriangle$: results for linear adduct formation, see Fig. 7, A_2/D replaced by C with $n=2$ and $\gamma_1=2.34$, or A_2/D replaced by A_1/D with $n=2$ and $\gamma_1=12.5$.

where n and γ_1 depend on θ , χ_S^2 , and χ_{AS}^2 . For the linear adduct, in Eq. (5.3) one has $n=2$, $\gamma_1=2.34$, and C instead of A_2/D while a factor of 1.5 in front of Z_2 instead of 0.75 in Eq. (5.2) gives a slightly better fit (replacing A_2/D by A_1/D instead of C , one has $n=2$ and $\gamma_1=12.5$). Figure 9 shows the quality of the fit with Eqs. (5.2) and (5.3) for a large number of nonlinear adduct systems. Examples for the fit parameters n and γ_1 from the CT calculations are given in Table IV. They can be reasonably well represented by

$n = 1 - 0.5 \sin^2 \theta + \sin^4 \theta$ (5.4)

and

$\gamma_1 = d^{-1/n}$ (5.5)

with

TABLE IV. Fit parameters n and γ_1 in Eq. (5.3) for low-temperature thermal rigidity factors: nonlinear adduct formation with potential of Eqs. (2.7)–(2.10).

$\theta(^{\circ})$	χ_S^2, χ_{AS}^2	n	γ_1
90	1,1	1.5	1.197
	1,4	1.5	0.622
	4,1	1.5	0.610
	4,4	1.5	0.306
60	1,1	1.1875	0.694
	1,4	1.1875	0.410
	4,1	1.1875	0.389
	4,4	1.1875	0.241
45	1,1	1.0	0.440
	1,4	1.0	0.266
	4,1	1.0	0.205
	4,4	1.0	0.149
30	1,1	0.9375	0.221
	1,4	0.9375	0.138
	4,1	0.9375	0.078
	4,4	0.9375	0.064

$$d = C_1 + C_2 \sin^2 \theta + C_3 / \sin^2 \theta + (\chi_S^2 \chi_{AS}^2)^{1/3} (C_4 + C_5 \sin^2 \theta + C_6 / \sin^2 \theta) + \chi_{AS}^2 \cos^2 \theta (C_7 + C_8 \sin^2 \theta + C_9 / \sin^2 \theta) + \chi_S^2 \cos^2 \theta (C_{10} + C_{11} \sin^2 \theta + C_{12} / \sin^2 \theta), \quad (5.6)$$

where $C_1 = 1.757$, $C_2 = -1.337$, $C_3 = -0.393$, $C_4 = 1.444$, $C_5 = -0.509$, $C_6 = -7.730$, $C_7 = -0.357$, $C_8 = -0.557$, $C_9 = 0.247$, $C_{10} = -1.932$, $C_{11} = 0.385$, and $C_{12} = 1.529$.

The high-temperature decay of f_{rigid} below $f_{\text{rigid}}(T \rightarrow 0)$ can be represented in the form

$$f_{\text{rigid}} = \delta_1 + \delta_2 \exp(X^{**}/\delta_3), \quad (5.7)$$

where $X^{**} = \ln(kT/D) + 4 - \beta r_e$. The fit parameters δ_2 are given in Table V. One may again construct a simplified general express analogous to Eq. (4.6).

The described calculations for nonlinear adducts all have been performed with the standard potential, i.e., $\alpha/\beta = 0.5$. Deviations from this situation ($\alpha/\beta \neq 0.5$) again can be treated as for linear adducts, i.e., using Eqs. (4.9) and (4.10) [with C and C_{eff} replaced by A_2/D and $(A_2/D)_{\text{eff}}$, respectively]; the derived $(A_2/D)_{\text{eff}}$ replace A_2/D in Eq. (5.3).

VI. GENERALIZED EXPRESSIONS FOR THERMAL RIGIDITY FACTORS

The investigations of the present work and of part IV¹ have shown a consistent pattern of thermal rigidity factors in those cases where the potential did not show too large anharmonicity and vibration-rotation coupling effects in the angular potential. In the following, we summarize our results for these simple cases. Using the general anisotropy parameter C defined by⁹

$$C = [\epsilon(r_e)]^2 / 2BD, \quad (6.1)$$

with the geometrical mean of the quanta of all transitional modes for $\epsilon(r_e)$ and the arithmetic mean of the corresponding rotational constants for B , all these cases are closely related.

At temperatures with $\ln(kT/D) < -6$, apart from the quantum effects treated in Sec. VII, the thermal rigidity factors for $\alpha/\beta = 0.5$ approach limiting values

$$f_{\text{rigid}}(T \rightarrow 0) \approx (1 + Z + Z^4)^{-1/4}, \quad (6.2)$$

where Z is related to C in different ways (optimum fit coefficients in front of Z vary between 0.75 and 1.5 but the differences are of little practical relevance). For case I, an atom adding to a linear rotor forming a T-shaped adduct, one has approximately

$$Z \approx C^{1/2} / \sqrt{6}; \quad (6.3)$$

for case II, an atom adding to a linear rotor forming a linear adduct, one has approximately

$$Z \approx C / \sqrt{2}; \quad (6.4)$$

for case III, a linear rotor adding to a linear rotor forming an adduct with parallel rotors (perpendicular to the line connecting the centers of mass), one has approximately

$$Z \approx C^{3/2} / \sqrt{12}; \quad (6.5)$$

for case IV, a linear rotor adding to a linear rotor forming a linear adduct, one has approximately

TABLE V. Fit parameters δ_i in Eq. (5.7) for thermal rigidity factors: non-linear adduct formation with potential of Eqs. (2.7)–(2.10).

$\theta(^{\circ})$	χ_S^2, χ_{AS}^2	A_2/D	δ_1	δ_2	δ_3
90	1,1	1.0	0.8631	−1.097	1.664
		2.0	0.4329	−0.552	2.347
		4.0	0.1372	−0.194	1.929
	1,4	1.0	0.6442	−0.895	2.049
		2.0	0.2126	−0.311	1.972
		4.0	0.0729	−0.096	2.248
	4,4	0.5	0.7681	−1.116	1.729
		1.1	0.2784	−0.407	1.992
		2.4	0.0766	−0.114	1.859
	60	1,1	0.7101	−0.782	1.910
		2.2	0.2789	−0.411	1.782
		5.0	0.0862	−0.120	2.041
45	1,4	0.7	0.6302	−0.576	2.291
		1.6	0.2485	−0.316	2.097
		3.5	0.0790	−0.118	1.941
	4,1	0.6	0.5997	−0.611	1.749
		1.3	0.3162	−0.410	2.027
		3.0	0.0944	−0.141	1.886
	4,4	0.3	0.7511	−0.520	2.436
		0.7	0.3841	−0.515	1.726
		1.5	0.1600	−0.221	2.178
	1,1	0.7	0.6506	−0.502	2.670
		1.5	0.2964	−0.309	2.184
		3.0	0.1283	−0.182	1.706
	1,4	0.5	0.5390	−0.438	2.302
		1.0	0.2700	−0.296	1.978
		2.0	0.1209	−0.146	1.905
	4,1	0.2	0.7304	−0.393	2.607
		0.5	0.3788	−0.323	1.705
		1.2	0.1928	−0.220	1.886
30	4,4	0.12	0.8092	−0.638	1.608
		0.27	0.4866	−0.394	1.951
		0.9	0.1769	−0.198	1.779
	1,1	0.3	0.6647	−0.498	2.389
		0.6	0.3492	−0.292	2.128
		1.3	0.1610	−0.157	2.120
	1,4	0.15	0.7504	−0.695	1.704
		0.3	0.4168	−0.370	2.016
		0.9	0.1414	−0.123	2.274
	4,1	0.09	0.6503	−0.418	2.448
		0.2	0.3329	−0.247	1.907
		0.5	0.1605	−0.102	2.563
	4,4	0.09	0.5640	−0.368	2.831
		0.2	0.2764	−0.226	1.978
		0.5	0.1267	−0.093	2.428

$$Z \approx C^2/2. \quad (6.6)$$

The exponent of C denotes one-half of the number of transitional modes, i.e., of rotors transforming into adduct oscillator.

There are two types of temperature dependences entering f_{rigid} . The repulsive part of the potential causes a high-temperature decrease of f_{rigid} below $f_{\text{rigid}}(T \rightarrow 0)$ from Eq. (6.2). In part IV,¹ e.g., for case II and $0 \leq C \leq 2$,

$$f_{\text{rigid}}/f_{\text{rigid}}(T \rightarrow 0) \approx 1 - 0.47C \exp(X/2.044) \quad (6.7)$$

with $X = \ln\{kT/[D \exp(\beta r_e)]\}$ providing a reasonable fit while in the present work, e.g., for case IV and $0 \leq C \sqrt{\beta r_e} \leq 7$,

$$f_{\text{rigid}}/f_{\text{rigid}}(T \rightarrow 0) \approx 1 - 2.3C \sqrt{\beta r_e} \exp(X/2.044) \quad (6.8)$$

the same X worked well. With the tabulated parameters, analogous expressions can be constructed for the other cases. A second temperature dependence of f_{rigid} arises when the ratio of the looseness parameter α and the Morse parameter β differs from the "standard value" $\alpha/\beta = 0.5$. We could show that this effect can be well accounted for by replacing C by an effective value C_{eff} , which is related to C through

$$C_{\text{eff}} = CF(kT/D)^{2\alpha/\beta - 1} \quad (6.9)$$

with different F for different cases. For case II, e.g., one has¹

$$F \approx 1 + 0.9(2\alpha/\beta - 1)^2 - 0.8(2\alpha/\beta - 1)^3, \quad (6.10)$$

while the present work given for case IV gave

$$F \approx 1 + 0.4(2\alpha/\beta - 1) + (2\alpha/\beta - 1)^2, \quad (6.11)$$

for $0.3 \leq \alpha/\beta \leq 0.7$, the various fits of F for the various cases did not differ too much.

The relation between Z , C , and the number of transitional modes b , expressed by $Z \propto C^{b/2}$, was derived earlier⁹ using simplified SACM oscillator models; however, the correct numerical relationships (6.3)–(6.6) required the CT treatment of the present work. Equations (6.1)–(6.6) appear most useful as an orientation in practical applications. However, stronger anharmonicity and vibration–rotation coupling effects complicate the situation when nonlinear or nonparallel rotor arrangements in the adducts are formed. We illustrate this in the following by comparing cases I and II with intermediate cases in Fig. 10, and cases III and IV with intermediate cases in Fig. 11. One notices that with increasing C the intermediate cases approach case I or III, respectively, i.e., the case of T-shaped or parallel-rotor adducts.

VII. SACM TREATMENT OF LOW-TEMPERATURE RIGIDITY FACTORS AND LOCAL NONADIABATICITIES

The results from the CT calculations of the previous sections have provided analytical approximate expressions for thermal capture rate constants and the corresponding thermal rigidity factors in the range where classical mechanics applies. In addition, the transition from globally adiabatic to globally nonadiabatic dynamics was demonstrated. In the present section we provide quantum SACM calculations for the range where classical mechanics breaks down. We also show that, with increasing temperature, the rigorous SACM

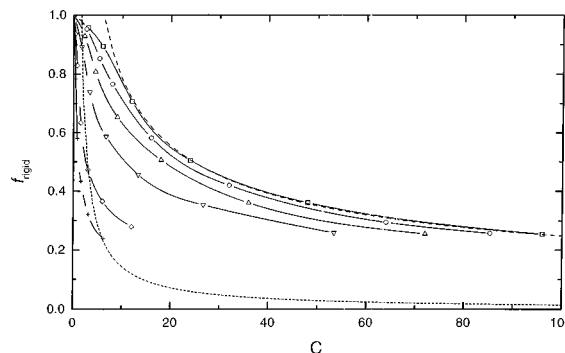


FIG. 10. Low-temperature thermal rigidity factors for the capture of atoms by linear rotors leading to nonlinear adducts. C = anisotropy parameter from Eq. (6.1) [CT results from part IV (Ref. 1); lines without points: Eq. (6.2) with Eq. (6.3) for T-shaped adducts (dashed) or with Eq. (6.4) for linear adducts (dotted); lines with points for $\gamma_e = 90^\circ$ ($\square$), 70.5° ($\circ$), 60° ($\triangle$), 48.2° (∇), 30° ($\diamond$), and 21° (+) where γ_e is the angle between the rotor axis and the line connecting the centers of mass of the rotor and the atom].

treatment approaches the adiabatic CT results. In our treatment we account for individual avoided crossings of adiabatic channel potential curves by considering the splittings. We also provide empirical average corrections for avoided crossings and nonadiabatic dynamics to be applied in harmonic oscillator versions of SACM.

We have at first calculated individual adiabatic channel potential curves for the potential of Eq. (2.1), i.e., for linear adduct formation with $\alpha/\beta = 0.5$. Channel potential curves were obtained by matrix diagonalization of the Hamiltonian such as described in Refs. 6–8. The treatment indicates that the six lowest channels do not show avoided crossings. However, higher channels are characterized by increasing numbers of simple or multiple avoided crossings between pairs or groups of channels. At all of these avoided crossings channel-hopping may occur and suitable branching ratios for the dynamics have to be calculated.^{6,8,11,12} With increasing energy this becomes a hopelessly laborious task such that the SACM treatment has to be replaced by a CT treatment, similar to the situation in the range of global nonadiabaticity. Nevertheless, the SACM treatment can be well extended into

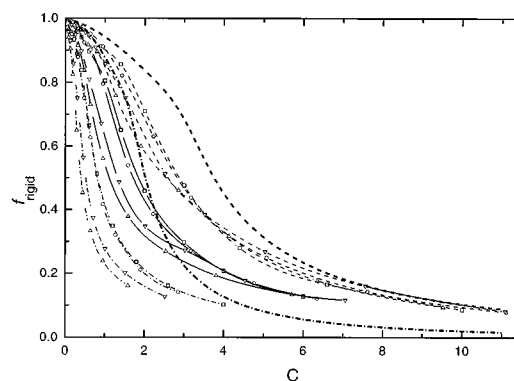


FIG. 11. Low-temperature thermal rigidity factors as a function of the anisotropy parameter C from Eq. (6.1) [CT results from this work; lines without points: Eq. (6.2) with Eq. (6.5) for parallel rotors (dashed) or with Eq. (6.6) for linear rotors in the adduct (dash-dotted); lines with points: $\{\chi_s^2, \chi_{AS}^2\} = \{1,1\}$ ($\square$), $\{1,4\}$ ($\circ$), $\{4,1\}$ ($\triangle$), $d\{4,4\}$ (∇), $\theta = 60^\circ$ (dashed), $\theta = 45^\circ$ (full), $\theta = 30^\circ$ (dash-dotted)].

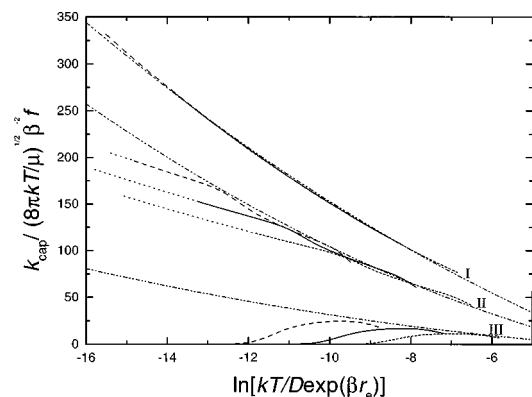


FIG. 12. Low-temperature capture rate constants from SACM calculations for the potential of Eq. (2.1) with $\alpha/\beta=0.5$ ($\beta r_e=4$; groups of curves I: $A_1/D=2$, II: $A_1/D=4$, low-temperature end from SACM perturbation treatment, III: $A_1/D=8$; dotted curves: $A_1/B=500$, full curves: $A_1/B=2000$, dashed curves: $A_1/B=10\,000$; fully adiabatic dynamics for splittings $\Delta E > 0.1B$ at avoided crossings and fully diabatic dynamics for $\Delta E < 0.1B$; dash-dotted lines: CT calculations).

the classical range when suitable channel-hopping criteria are employed: we assumed complete channel-hopping when the calculated splitting of the adiabatic-channels at the Landau-Zener-type avoided crossings is smaller than a certain fraction of B , or no hopping when the splitting is larger. Errors due to channel branching are largely averaged out in this treatment of thermal capture rate constants; however, product state distributions in unimolecular bond fissions are not expected to be well reproduced.

Assuming fully diabatic dynamics (i.e., without following the true adiabatic channel potentials which avoid crossing) when the splitting ΔE at avoided crossings is smaller than $0.1B$ and fully adiabatic dynamics for $\Delta E \geq 0.1B$ (i.e., following the avoided crossing), thermal capture rate constants from rigorous SACM and CT calculations are presented in Fig. 12. A first group (I) of SACM calculations with $A_1/D=2$ (corresponding to $C=\sqrt{3}/2=0.866$) over wide temperature ranges gives practically identical results as the CT calculations described in Sec. IV. About 600 channels were included (corresponding to rotational quantum numbers up to $j=4$ in each linear rotor). In a second group (II) of calculations with $A_1/D=4$ ($C=\sqrt{3}=1.732$), the SACM results with increasing temperature approach the CT results from below, show some structure due to the opening of higher channels, partly overshoot, and finally merge with the CT results (at the highest temperatures the shown SACM results start falling below the CT results because of the neglect of omitted higher channels). In a third group (III) with $A_1/D=8$ ($C=2\sqrt{3}=3.464$), $k_{\text{cap}}^{\text{SACM}}$ approaches zero for $T \rightarrow 0$. The situation is quite comparable to the addition of atoms to linear rotors, see part IV.¹ The difference between groups I and II and group III is due to the appearance of adiabatic zero-point barriers for $C > \sqrt{3}$ ($A_1/D > 4$) whereas no such barriers arise for $C \leq \sqrt{3}$.

The described quantum effects occur at temperatures which are roughly of the order of $kT \approx B$. Figure 13 shows the ratio $k_{\text{cap}}^{\text{SACM}}/k_{\text{cap}}^{\text{CT}}$ as a function of $\ln(kT/B)$. The effects of increasing zero-point barriers with increasing C are quite

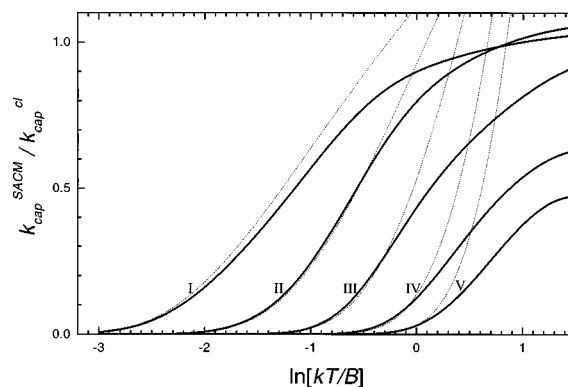


FIG. 13. Same as Fig. 12, reduced representation of capture rate constants from SACM ($k_{\text{cap}}^{\text{SACM}}$) and CT ($k_{\text{cap}}^{\text{CT}}$) [$\beta r_e=4$; $A_1/B=10\,000$; curves I: $A_1/D=5$, II: $A_1/D=6$, III: $A_1/D=8$, IV: $A_1/D=12$, V: $A_1/D=16$, full lines: SACM, dotted curves: low-temperature limits from Eqs. (7.1)–(7.6)].

evident. As long as there are adiabatic zero-point barriers, i.e., when the lowest channel potential for capture has a potential maximum E_0 ($l=0$) larger than zero, the barriers E_0 ($l=0$) (l =quantum number of orbital rotation) dominate the “activated complex position” $r^\ddagger$ of the channel maxima for all l . With $E_0(l=0)$ and $r^\ddagger$ for the lowest channel one may well approximate $k_{\text{cap}}^{\text{SACM}}$ in the low-temperature limit (see Ref. 1). With the explicit channel potential from Ref. 6, we have calculated and represented $E_0(l)$ and $r^\ddagger$. Our results, for $\sqrt{3} \leq C \leq 3$, are

$$E_0(l=0)/B \approx 6(1-\sqrt{3}/C)^2 + 0.0844(C-\sqrt{3})^3, \quad (7.1)$$

$$a^\ddagger \approx 6(1-\sqrt{3}/C) + 0.593(C-\sqrt{3})^{2.5}, \quad (7.2)$$

and, for $C \geq 3$,

$$E_0(l=0)/B \approx (\sqrt{3}+1)^2 C / (4\sqrt{3}) - 2.60/C^{1/4}, \quad (7.3)$$

$$a^\ddagger \approx (\sqrt{3}+1)^2 C^2 / 24, \quad (7.4)$$

where

$$a^\ddagger = (D/B)(C/\sqrt{3}) \exp[-\beta(r^\ddagger - r_e)]. \quad (7.5)$$

With these $E_0(l=0)$ and $r^\ddagger$, the limiting low-temperature rate constants are approximated by¹

$$k_{\text{cap}}^{\text{SACM}}(T \rightarrow 0) \approx f r^{\ddagger 2} \sqrt{8\pi kT/\mu} \exp[-E_0(l=0)/kT]. \quad (7.6)$$

Figure 13 includes the corresponding rate constants which well reproduce the limiting results from the complete SACM treatment. Because $k_{\text{cap}}^{\text{SACM}}$ does not differ from $k_{\text{cap}}^{\text{CT}}$ too much over the range of practical interest for $C \leq \sqrt{3}$, where no adiabatic zero-point barriers are found, we do not further model quantum low-temperature behavior in this case.

Nonadiabaticity effects disappear in the low-temperature limit when only channels contribute which do not show avoided crossings. The six lowest channels are of this type, see Ref. 6. With increasing temperature, the effects of avoided crossings become relevant. The corresponding dynamics may effectively lower the adiabatic channel barriers, constructed neglecting avoided crossings, by populating lower channels at avoided crossings. The branching ratio in general will depend on the velocity of passing the crossing

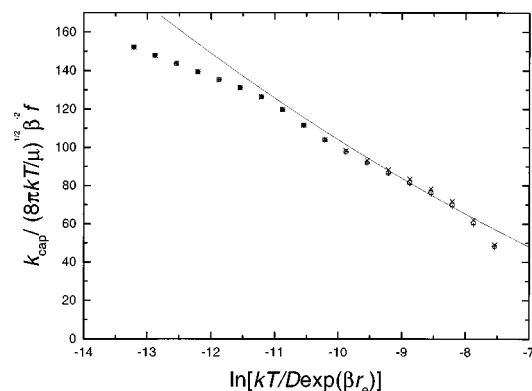


FIG. 14. Same as Fig. 12, for $A_1/D=4$, $A_1/B=2000$, $\beta r_e=4$. Results for different splitting criteria $\Delta E=0$ ($\times$), $\Delta E>0.1B$ ($\circ$), and $\Delta E>B$ ($+$), see the text; full line: CT results.

area.^{6,8,11,12} In order to obtain at least an estimate of the effects on thermal capture rates, we have at first calculated $k_{\text{cap}}^{\text{SACM}}$ for fully adiabatic behavior ($\Delta E=0$). Then we have calculated $k_{\text{cap}}^{\text{SACM}}$ following the adiabatic potentials only when the splitting ΔE at avoided crossings is larger than $0.1B$ ($\Delta E \geq 0.1B$), and finally when the splitting is larger than B ($\Delta E \geq B$). Crossings with smaller splittings were assumed to be characterized by fully diabatic dynamics, i.e., by complete channel hopping which leads to effectively higher channel maxima. As a consequence $k_{\text{cap}}^{\text{SACM}}$ will decrease in the order of $\Delta E=0$, $\geq 0.1B$, $\geq B$. For example, we have chosen CT calculations and SACM calculations for the case $A_1/D=4$ ($C=\sqrt{3}$), $\beta r_e=4$, $A_1/B=2000$. Figure 14 shows the results. As mentioned above, the S-shaped form of $k_{\text{cap}}^{\text{SACM}}$ is no artifact; it is real, reflecting the channel pattern with contributions from few attractive and repulsive potentials. At low temperatures the splitting criterion does not matter, see above; at higher temperatures $k_{\text{cap}}^{\text{SACM}}$ for $\Delta E=0$ is about 5% higher than for $\Delta E \geq B$. (The drop of $k_{\text{cap}}^{\text{SACM}}$ at the highest temperatures is again a numerical artifact due to the limited number of included channels.)

To be fully rigorous on the SACM level, instead of the described splitting criterion one should have implemented velocity-dependent branching ratios. One might also have analyzed the crossing dynamics by local nonadiabaticity Massey criteria.⁶ Because of the enormous number of avoided crossings to be accounted for at moderately high temperatures, we have preferred to determine empirical correction factors in the following way. Harmonic oscillator models of SACM (like the original version of SACM from Ref. 5) do not show any avoided crossings of adiabatic channel potentials and consequently no local nonadiabaticity. These models, therefore, produce a certain fraction of adiabatic channel maxima which are too high. By comparison with CT calculations we found that these effects lead to thermal capture rate constants which are 8/9 of the true CT results in the classical range, see above. However, this empirical factor only applies to typical valence potentials with channel maxima in the range where the transitional modes are mostly of oscillator type (i.e., at shorter distances r). On the other hand, true dipole-dipole capture generally is characterized by channel maxima in a range where avoided

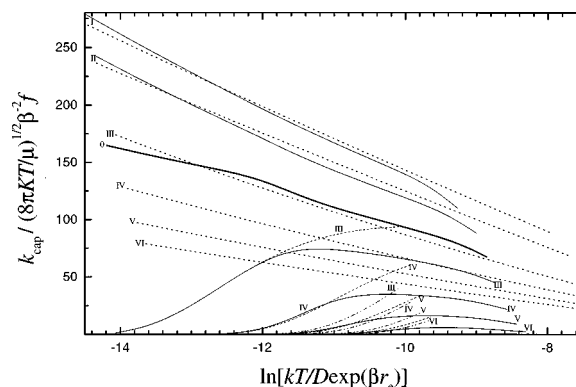


FIG. 15. Low-temperature capture rate constants from SACM (full lines) and CT (dotted lines) calculations for the potential of Eq. (2.7) with $\alpha/\beta=0.5$ [$\beta r_e=4$; parallel rotors in adduct, i.e., $\theta=90^\circ$; $\chi_S^2=\chi_{AS}^2=1$; $C=A_2/2D=0.5$ (I), 0.65 (II), 0.75 (0), 0.85 (III), 1 (IV), 1.15 (V), 1.3 (VI); $A_2/B=2000$; dashed lines: lowest channel capture, dash-dotted lines: lowest channel threshold capture].

crossings between hindered rotor-type potentials is more frequent. Here thermal capture rate constants were found to be about 8/10 of the true CT results.⁴ For the present case of valence potentials, we recommend the use of an empirical avoided-crossing and nonadiabaticity correction factor of 9/8 to be employed in high-temperature harmonic oscillator versions of SACM. For low temperatures, the true 26 lowest adiabatic channel potentials from Ref. 6 may be adopted and combined with harmonic oscillator potentials for higher channels. Alternatively, SACM/CT transition curves may be constructed with the low-temperature limits given by Eqs. (7.1)–(7.6) and the transition curves from Fig. 13.

One should keep in mind that low-temperature capture on the SACM level can also account for open electronic shells effects. For example, in Ref. 13 we have treated the low-temperature combination of two HO radicals in their $^2\Pi_{3/2}$ electronic states.

Finally we describe rigorous SACM calculations for the potential of Eq. (2.7) with $\theta=\pi/2$, i.e., with parallel rotors in the adduct. Figure 15 shows comparisons of SACM and CT capture rate constants. The same qualitative behavior is observed as in Fig. 12: for weak anisotropies of the potential (curves I, II, and 0, $C=A_2/2D \leq 0.75$), no adiabatic zero-point barriers arise and SACM and CT rate constants are close to each other (no CT curve is shown for curve 0); for stronger anisotropies (curves III–VI, $C \geq 0.85$), adiabatic zero-point barriers $E_0(l=0)$ arise which are different from zero and which force $k_{\text{cap}}^{\text{SACM}}(T \rightarrow 0)$ towards zero. For the curves III–VI, Fig. 15 includes SACM capture rate constants on the lowest adiabatic channel which well represent the low-temperature limit. Fixing the adiabatic channel maxima $E_0(l)$ to the position of the channel maximum $E_0(l=0)$ of the lowest channel, gives insufficient results at medium anisotropies (curve III) while the agreement with complete SACM calculations improves with increasing anisotropy (curve VI). The SACM calculations in Fig. 15, like those in Figs. 12–14, have included about 600 channels (up to $j=4$ in each rotor). This limitation in the number of channels causes the decrease of $k_{\text{cap}}^{\text{SACM}}$ at $\ln[kT/D \exp(\beta r_e)] \geq -10$ which thus is a calculational artifact. The stronger the anisot-

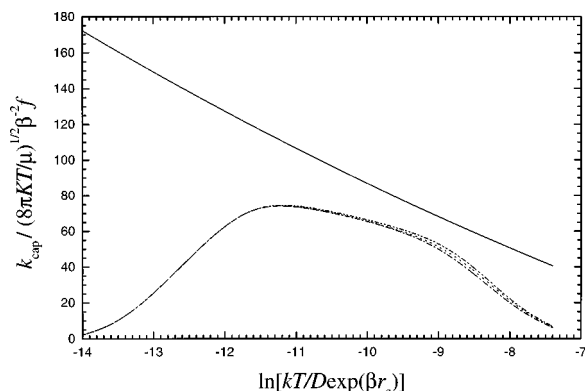


FIG. 16. Same as Fig. 15, for $C=A_2/2D=0.85$ (III). Full line: CT; dotted line: SACM with diabatic behavior for $\Delta E \leq 0.1B$, dash-dot-dotted line: SACM with diabatic behavior for $\Delta E \leq B$, dash-dotted line: SACM with completely adiabatic behavior.

ropy, the less well the classical $k_{\text{cap}}^{\text{cl}}$ is approached with the included number of channels (curves III–VI). Apart from a factor $\exp(-E_0(l=0)/kT)$, apparently $k_{\text{cap}}^{\text{SACM}}$ approaches $k_{\text{cap}}^{\text{cl}}$ as a linear function of $\ln[kT/D]$ until the curves intersect and then follows $k_{\text{cap}}^{\text{cl}}$ (similar to curve 0).

Finally we also demonstrate the effects of curve-hopping at Landau–Zener-type crossing points in the case of nonlinear adduct formation. We have chosen curve III from Fig. 15 as an example. The completely adiabatic treatment, following all potentials at their avoided crossings, leads to the lowest channel maxima and, hence, to the largest rate constant, while the diabatic treatment for all channels with channel crossings $\Delta E \leq B$ leads to the smallest rate constants. Figure 16 shows the effect for these two cases and the intermediate case $\Delta E \leq 0.1B$ (the decay at $\ln[kT/D \exp(\beta r_e)] \geq -9$ is the mentioned calculational artifact). The nonadiabaticity effects in Fig. 16 are comparable in size to those of Fig. 14. However, it should be mentioned that, in the last case of adducts with parallel rotors (perpendicular to the line connecting the centers of mass), the approach of $k_{\text{cap}}^{\text{SACM}}$ to $k_{\text{cap}}^{\text{cl}}$ extends up to much higher temperatures than in the case of linear adducts.

VIII. CONCLUSIONS

The present work in Sec. VI has provided explicit relationships between thermal rigidity factors and the general anisotropy parameter C which is related to molecular prop-

erties in a transparent way. The earlier work from Refs. 1 and 9 could be extended to more complex capture cases. However, simple expressions were only found for linear, T-shaped, or parallel rotor arrangements in the adduct where vibrations and rotations can well be separated. Other nonlinear adducts were also treated explicitly, however, leading to more complicated thermal rigidity factors. The present relationships can easily be used in practice. However, they are based on standardized valence potentials. Nevertheless, modifications can be applied in a rational way when more details of the potentials are known.

A characteristic difference between the capture of linear rotors by atoms or by linear rotors was identified. The former shows only rare channel pseudocrossings, whereas the latter (except at low energies) is dominated by such crossings. The CT treatment provides an empirical quantification of the avoided crossing and nonadiabaticity contributions to harmonic oscillator SACM rate constants. In summary, the combined SACM/CT treatment provides the most rigorous, explicitly tractable approach to capture processes over wide temperature ranges available today.

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